

Erdős Problem #282

EP-282	L1 - Tractable	Ouvert	Number Theory
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Énoncé original (anglais, reproduction mot pour mot)

Let $A \subseteqq \mathbb{N}$ be an infinite set and consider the following greedy algorithm for a rational $x \in (0, 1)$: choose the minimal $n \in A$ such that $n \geq 1/x$ and repeat with x replaced by $x - \frac{1}{n}$. If this terminates after finitely many steps then this produces a representation of x as the sum of distinct unit fractions with denominators from A . Does this process always terminate if x has odd denominator and A is the set of odd numbers? More generally, for which pairs x and A does this process terminate?

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